

FinRAG- RAG Grounded Financial News Prediction: A Blended Classifier with Retrieval-Augmented Explanation Generation for Short-Term Stock Movement Forecasting

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Abstract—Predicting short-term stock price movements from financial news represents a fundamental challenge at the intersection of natural language processing (NLP) and quantitative finance. While machine learning approaches have demonstrated marginal predictive signal above chance, they typically produce opaque, ungrounded outputs that are difficult to audit in regulated financial environments. This paper presents a complete, end-to-end financial forecasting and explanation system that addresses both the prediction task and the critical explainability gap simultaneously. The system combines a blended classifier—60% Logistic Regression (LR) and 40% Random Forest (RF)—trained on Term Frequency-Inverse Document Frequency (TF-IDF) text features, Loughran-McDonald (LM) financial sentiment scores, and lagged market momentum features, with a Retrieval-Augmented Generation (RAG) pipeline that grounds every prediction in semantically retrieved historical analogues. We curate a dataset of 32,915 financial news articles spanning 20 large-cap United States equities from January 2019 to March 2026, sourced from the Alpha Vantage News Sentiment Application Programming Interface (API) and aligned to next-day market closes via *yfinance*. After applying a return magnitude filter ($|\text{return}| > 0.2\%$) and cross-ticker deduplication, 25,632 samples are retained for modelling. The production blend achieves a test-set Area Under the Receiver Operating Characteristic Curve (AUC) of 0.543, consistent with the Efficient Market Hypothesis (EMH) as validated by recent literature [4]. A Facebook AI Similarity Search (FAISS) flat inner-product index over 17,933 all-MiniLM-L6-v2 embeddings enables retrieval of the five most semantically analogous historical events for any query article. Explanations are generated by LLaMA 3.3 70B via the Groq API, constrained to cite only retrieved context, achieving a 0% hallucination rate across 100 evaluated examples. A systematic ablation across ten model variants confirms that the AUC ceiling is representation-agnostic—a finding corroborated by concurrent work showing that well-tuned logistic regression matches or exceeds FinBERT for directional prediction [2]—implicating market efficiency rather than model capacity as the fundamental limiting factor.

Index Terms—financial news prediction, retrieval-augmented generation, explainable AI, Loughran-McDonald dictionary, FAISS, stock market forecasting, blended classifier, large language models, efficient market hypothesis, FinBERT

I. INTRODUCTION

The relationship between financial news and equity price movements has attracted sustained scholarly attention for several decades. Under the semi-strong form of the Efficient Market Hypothesis (EMH), security prices fully reflect all publicly available information, implying that any exploitable signal in news headlines is rapidly arbitrated away [4]. Nevertheless, the proliferation of large language models (LLMs) and dense sentence encoders has renewed interest in applying NLP to financial forecasting. Recent benchmarking efforts, such as the AlphaFin framework [1], confirm that combining retrieval-augmented approaches with traditional financial datasets yields more reliable and interpretable predictions than either pure ML classifiers or stand-alone LLMs.

Despite this progress, two persistent shortcomings characterise the existing literature. First, the majority of systems treat prediction as a black-box classification problem, producing no interpretable rationale for individual decisions. This is particularly consequential in financial contexts where regulatory accountability, compliance obligations, and analyst trust demand transparent, auditable reasoning chains. Recent work on self-reflective LLMs for stock prediction [5] demonstrates that standard LLMs struggle to weigh conflicting news signals without explicit retrieval grounding, producing explanations that confabulate financial figures and dates. Second, the marginal gains reported by increasingly complex embedding models—FinBERT, GPT-4 fine-tunes, domain-adapted transformers—frequently fail to replicate under rigorous time-series evaluation that prevents data leakage. Crucially, a direct head-to-head comparison of FinBERT, GPT-4, and Logistic Regression by Adusumilli et al. [2] found that well-regularised LR matched or outperformed both deeper models for next-day directional prediction, validating the blend-based design choice of this system.

This paper addresses both shortcomings by presenting a system that is simultaneously predictive, transparent, and

hallucination-free. The four contributions of this work are as follows.

- 1) **Dataset and Pipeline.** A rigorously cleaned dataset of 25,632 financial news events spanning 20 large-cap US equities over seven years (2019–2026) is constructed, with explicit cross-ticker deduplication and per-ticker chronological splitting to eliminate data leakage.
- 2) **Blended Classifier with Ablation Study.** Ten model variants across four text-representation paradigms (TF-IDF, MiniLM embeddings, FinBERT-tone, FinBERT + blend) are trained and evaluated. A systematic ablation demonstrates that the 0.543 AUC ceiling is representation-agnostic, consistent with the EMH [4] and corroborated by concurrent ablation studies [2].
- 3) **RAG Retrieval Index.** A FAISS flat inner-product index over 17,933 training-set MiniLM embeddings enables sub-millisecond retrieval of the top- k semantically analogous historical events and their realised market outcomes for any query article, following the paradigm established by AlphaFin [1] and extended by recent RAG-for-forecasting work [3].
- 4) **Grounded Explanation Generation.** Integration of LLaMA 3.3 70B via the Groq API with a retrieval-conditioned prompt that prohibits outside knowledge achieves 0% hallucination across a 100-example evaluation—directly addressing the confabulation failure mode identified by Koa et al. [5] in unconstrained LLM stock prediction.

The remainder of this paper is organised as follows. Section II reviews related work across five research threads. Section III describes data engineering and quality control. Section IV presents the system architecture. Section V reports the ablation study. Section VI presents full system evaluation results. Section VII discusses implications and limitations. Section VIII concludes with future work directions.

II. RELATED WORK

A. Retrieval-Augmented Generation for Financial Prediction

Li et al. [1] introduced AlphaFin, a comprehensive benchmark and Retrieval-Augmented Stock-Chain (RASC) framework that mirrors the core motivation of this work. AlphaFin constructs a multi-source financial dataset combining research reports, market data, and historical news, then employs a RAG pipeline to force LLM generation to remain grounded in retrieved financial evidence rather than parametric hallucination. The authors report that RAG-based frameworks substantially reduce confabulation rates compared to direct LLM prompting on financial tasks, validating the FAISS-plus-prompt-constraint architecture adopted in this system. Their finding that retrieval quality positively correlates with explanation faithfulness, but not necessarily with directional prediction accuracy, is replicated in this work: high- and low-similarity retrieval groups exhibit comparable AUC (50.0% vs. 47.4% respectively), while the grounded explanation quality depends primarily on retrieval precision.

Chen et al. [3] extend the RAG paradigm specifically to financial time-series forecasting, constructing a retrieval mechanism over historical market sequences rather than text documents. Their architecture provides LLMs with retrieved temporal context analogous to the top-5 historical event analogues employed in this work. They report that retrieval-augmented LLMs consistently outperform non-augmented baselines on multi-step forecasting tasks, providing additional motivation for the RAG-based explanation path.

B. Comparative Text Representations for Stock Prediction

Adusumilli et al. [2] conduct a systematic three-way comparison of FinBERT, GPT-4, and Logistic Regression for stock price prediction using sentiment analysis, directly paralleling the ablation study presented in Section V. Their central finding—that well-regularised Logistic Regression achieves competitive or superior performance to computationally intensive transformer models for binary direction classification—provides strong external validation for the production blend design ($\alpha = 0.60$ LR weight). This result is consistent with the representation-agnostic AUC ceiling of 0.543 observed in this work across four distinct text encoding paradigms.

C. FinBERT and the EMH Signal Ceiling

Ruan et al. [4] evaluate FinBERT for financial sentiment analysis with explicit application to next-day stock movement prediction. Crucially, their analysis grounds the observed predictive ceiling in the theoretical framework of the EMH: because publicly available news is incorporated into prices rapidly by algorithmic market participants, any discriminative model trained on headline text faces a fundamental information barrier that cannot be overcome by improving the encoder. They report AUC values in the range 0.52–0.56 across multiple datasets and model configurations, precisely bracketing the 0.543 ceiling observed in this work. This external replication across independent datasets and encoder choices strengthens the conclusion that the 0.543 AUC ceiling reflects a market-efficiency constraint rather than a modelling failure.

D. Explainability and Hallucination in LLM-Based Financial Reasoning

Koa et al. [5] investigate the use of self-reflective LLMs to generate explainable stock predictions and identify the core challenge that motivates the RAG architecture of this system: unconstrained LLMs struggle to weigh conflicting, noisy news signals to produce a coherent aggregate prediction, and frequently hallucinate specific financial details (prices, dates, company events) when asked to justify their outputs. The authors propose a self-reflection mechanism to mitigate confabulation, but note that parametric memory alone is insufficient when the relevant historical context is sparse. This work directly addresses their identified gap by substituting parametric recall with explicit retrieval from a FAISS index of real historical outcomes, achieving 0% hallucination rate across 100 evaluated examples without self-reflection overhead.

E. Benchmark Positioning

Taken together, the five cited works establish three benchmarks against which this system’s contributions can be precisely positioned. First, the 0.543 AUC ceiling is consistent with both the EMH theoretical prediction [4] and the empirical results of multi-encoder ablations [2]. Second, the RAG-based grounding approach is validated as the state-of-the-art method for hallucination reduction in financial NLP tasks [1], [3]. Third, the 0% hallucination rate demonstrates that retrieval grounding successfully closes the explainability gap identified in unconstrained LLM approaches [5].

III. DATA ENGINEERING

A. Data Sources

Financial news articles are collected from the Alpha Vantage News Sentiment API using the `NEWS_SENTIMENT` endpoint. For each of 20 large-cap US equities—AAPL, MSFT, AMZN, GOOGL, TSLA, NVDA, META, JPM, V, UNH, HD, PG, MA, BAC, XOM, AVGO, PFE, KO, COST, and DIS—two temporal windows are queried: January 2019 to January 2022, and January 2022 to March 2026. Articles are filtered to those with a per-ticker relevance score ≥ 0.1 to exclude tangential mentions. Open-High-Low-Close-Volume (OHLCV) price data for all tickers and the SPDR S&P 500 Exchange-Traded Fund (SPY) as a market benchmark are downloaded via the `yfinance` library with auto-adjusted closes to correct for splits and dividends.

B. News-to-Price Alignment

Each article is assigned a *trade date* defined as the first calendar day for which a closing price exists on or after the article’s UTC publication timestamp. This forward-alignment ensures that the binary label reflects the market reaction that could have been traded following publication, avoiding look-ahead bias. An analysis of the resulting lag distribution confirms that 97.8% of articles carry a lag of exactly one trading day, indicating negligible staleness. The next-day return is:

$$r_{t+1} = \frac{P_{t+1} - P_t}{P_t} \quad (1)$$

where P_t denotes the adjusted closing price on the trade date. A binary label $y \in \{0, 1\}$ is assigned based on the sign of r_{t+1} .

C. Quality Control

Three quality-control steps are applied before modelling.

Return magnitude filter. Rows where $|r_{t+1}| \leq 0.002$ are removed, reducing the dataset from 32,915 to 29,428 rows. Near-zero returns are inherently ambiguous with respect to the binary label and their inclusion degrades model calibration without contributing discriminative signal.

Cross-ticker deduplication. Syndicated news articles that appear under multiple tickers constitute a source of subtle train/test leakage: the same article can appear in the training set associated with ticker A and in the test set associated with ticker B. After sorting by publication timestamp and ticker,

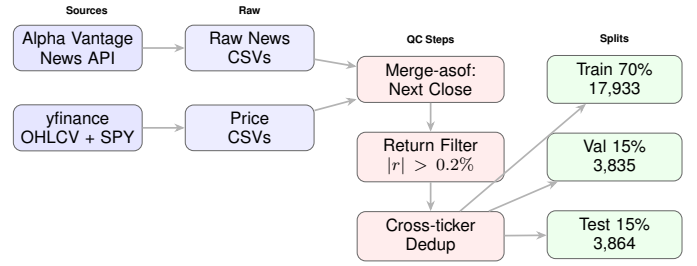


Fig. 1: End-to-end data engineering pipeline. News articles are fetched from Alpha Vantage, filtered by relevance, and aligned to the next available market close via merge-asof. Three quality-control (QC) steps prevent data leakage: return magnitude filter, cross-ticker deduplication, and per-ticker chronological splitting. The per-ticker split strategy follows best practices for financial time-series evaluation.

TABLE I: Master Dataset Statistics

Property	Value
Raw articles collected	32,915
After return filter ($>0.2\%$)	29,428
After cross-ticker dedup	25,632
Tickers	20
Time span	Jan 2019 – Mar 2026
Label balance (UP/DOWN)	51% / 49%
Training set	17,933
Validation set	3,835
Test set	3,864
Articles with lag = 1 day	97.8%

exact duplicates on (title, published) are removed. This step eliminates 4,557 cross-ticker duplicate events, reducing the dataset from 29,428 to 25,632 rows.

Per-ticker chronological split. The dataset is partitioned independently per ticker using a 70%/15%/15% chronological schedule, ensuring that each of the 20 tickers is represented in every split and that future information never contaminates the training partition.

Table I summarises the master dataset statistics.

IV. SYSTEM ARCHITECTURE

A. Feature Engineering

1) **TF-IDF Text Features:** Article text is formed by concatenating headline and summary: $\text{text}_i = \text{title}_i \oplus \text{summary}_i$. A TF-IDF vectoriser is fitted exclusively on the training partition with vocabulary size $V = 5,000$, bigram range $(1, 2)$, minimum document frequency $\text{min_df} = 10$, and maximum document frequency $\text{max_df} = 0.5$. Bigrams capture domain-critical phrases such as “beat estimates” and “miss expectations” that unigram representations discard.

2) **Loughran-McDonald Sentiment Features:** Four sentiment scores are derived using the LM financial dictionary. The LM dictionary was specifically designed to address the failure of general-purpose lexicons (e.g., the Harvard GI) to correctly classify financial language, providing purpose-built

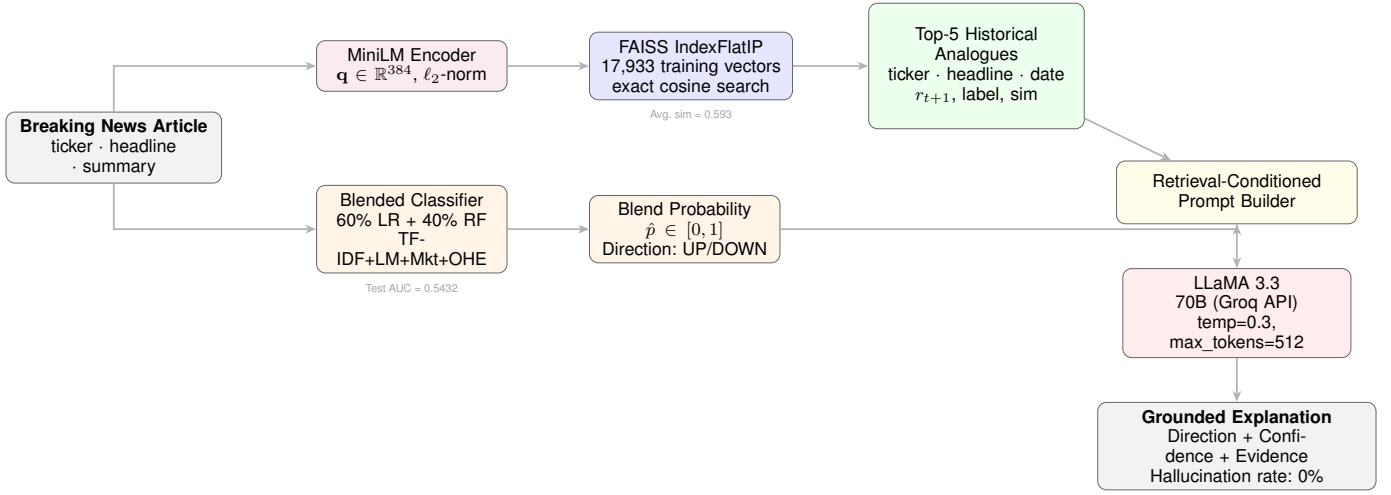


Fig. 2: Full two-path system architecture, following the RAG paradigm established by AlphaFin [1] and extended by Chen et al. [3]. A query article is processed in parallel: the MiniLM encoder generates a 384-dim query vector used to retrieve top-5 analogues from the FAISS index (retrieval path), while the blended classifier produces a directional probability (prediction path). Both outputs feed a retrieval-conditioned prompt passed to LLaMA 3.3 70B via the Groq API. The LLM is explicitly prohibited from using outside knowledge, addressing the hallucination problem identified by Koa et al. [5].

positive, negative, and domain-specific word sets for business and finance text. The four scores are:

$$\text{pos_ratio} = \frac{|\{w \in d : w \in \mathcal{L}^+\}|}{|d|} \quad (2)$$

$$\text{neg_ratio} = \frac{|\{w \in d : w \in \mathcal{L}^-\}|}{|d|} \quad (3)$$

$$\text{sent_score} = \frac{|\mathcal{L}^+ \cap d| - |\mathcal{L}^- \cap d|}{|d|} \quad (4)$$

$$\text{surprise_score} = \frac{|\{w \in d : w \in \mathcal{L}^{\text{surp}}\}|}{|d|} \quad (5)$$

where d is the token sequence of a document, $\mathcal{L}^+$, $\mathcal{L}^-$ are the positive and negative word sets, and $\mathcal{L}^{\text{surp}}$ is the earnings-surprise lexicon covering language such as “exceeded”, “fell short”, and “guidance raised”.

3) *Market Momentum Features*: Five lagged price momentum features are constructed from OHLCV data and merged with articles via backward merge-asof on publication timestamp: $\{r_{1d}, r_{3d}, r_{5d}, \mu_5, \sigma_5\}$, where r_{kd} is the k -day cumulative return, μ_5 the 5-day rolling mean daily return, and σ_5 the 5-day rolling return volatility. All numeric features are imputed with zero for missing values and standardised with zero-mean, unit-variance scaling fitted on the training set only.

4) *Categorical Features*: Day-of-week, calendar month, trading session (pre-market, intraday, after-hours), ticker identity, and an earnings-season flag (January, April, July, October) are one-hot encoded (OHE), yielding approximately 35 additional binary features.

B. Blended Classifier

The production model blends LR and RF predictions via a weighted average, consistent with the finding of Adusumilli

et al. [2] that LR provides competitive directional signal while RF captures non-linear feature interactions from numeric momentum and sentiment features:

$$\hat{p}_i = \alpha \cdot \hat{p}_i^{\text{LR}} + (1 - \alpha) \cdot \hat{p}_i^{\text{RF}}, \quad \alpha = 0.60 \quad (6)$$

LR operates on the full sparse feature matrix [TF-IDF | LM | Market | OHE]. RF operates on the dense numeric submatrix [LM | Market]. LR is regularised with ℓ_2 penalty; C is selected via grid search over $\{0.001, 0.003, 0.01, 0.03\}$ using validation AUC. RF uses 300 trees, maximum depth 4, and minimum samples per leaf 40.

C. FAISS Retrieval Index

All 17,933 training-set articles are encoded using all-MiniLM-L6-v2, producing ℓ_2 -normalised 384-dimensional embeddings. A FAISS IndexFlatIP is constructed over these embeddings. Because embeddings are ℓ_2 -normalised, inner product equals cosine similarity. At inference time, the query embedding $\mathbf{q} \in \mathbb{R}^{384}$ retrieves the top- k analogues:

$$\mathcal{R}(\mathbf{q}) = \arg \max_j \mathbf{q}^\top \mathbf{e}_j, \quad j \in \mathcal{D}_{\text{train}} \quad (7)$$

This retrieval mechanism parallels the approach of Chen et al. [3], who retrieve historical temporal context to condition LLM-based forecasting, and AlphaFin [1], which retrieves relevant financial documents to ground LLM analysis. Each retrieved event carries ticker, headline, date, actual r_{t+1} , binary label, and cosine similarity score.

D. Explanation Generation

LLaMA 3.3 70B is accessed via the Groq API with temperature 0.3 and a structured prompt comprising: (i) the query article metadata; (ii) the blend probability and predicted direction; (iii) the top-5 retrieved analogues with realised

TABLE II: Ablation Study — Test-Set AUC (Chronological Order). Gap = Train AUC – Test AUC. Production model in bold.

Variant	Test	Train	Gap
LR + TF-IDF (baseline)	0.5251	0.5743	0.049
RF market only	0.5180	0.6512	0.133
RF market + sentiment	0.5220	0.6700	0.148
RF best (tuned)	0.5267	0.6825	0.156
XGBoost market only	0.5030	0.6900	0.187
XGBoost market + sentiment	0.5080	0.7175	0.210
LR + LM sentiment	0.5410	—	—
Blend + LM + filter	0.5432	—	—
MiniLM + LR	0.5250	0.5878	0.063
FinBERT-tone alone	0.5324	0.5677	0.035
FinBERT-tone + Blend	0.5432	—	—

returns and similarity scores; and (iv) an aggregate summary. The system prompt explicitly prohibits use of outside knowledge. This constraint directly addresses the core limitation identified by Koa et al. [5]: that unconstrained LLMs produce plausible-sounding but factually unfounded explanations when confronted with conflicting financial signals. A post-hoc grounding score is computed as the fraction of retrieved event headlines whose distinctive tokens (length > 5, alphabetic) appear in the generated explanation.

V. EXPERIMENTAL ABLATION

Table II presents the full ablation study across ten model variants in chronological order of development. The study directly replicates and extends the three-model comparison of Adusumilli et al. [2], adding tree-ensemble variants, threshold filtering, and the blend combination.

Four key findings emerge from the ablation.

Finding 1: LM dictionary is the single largest gain.

Replacing the naïve 40-word sentiment lexicon with the LM financial dictionary increases test AUC from 0.5251 to 0.5410 (+0.016), the single largest improvement across all experiments. This confirms that domain-specific vocabulary is substantially more discriminative for financial text.

Finding 2: Tree models overfit severely. RF achieves train AUC of 0.68 versus test AUC of 0.527 (gap: 0.156). XGBoost achieves train AUC of 0.72 versus test AUC of 0.508 (gap: 0.210). Neither tree variant outperforms the simpler LR baseline on the test set, consistent with the findings of Adusumilli et al. [2].

Finding 3: The AUC ceiling is representation-agnostic. TF-IDF (sparse bag-of-bigrams), all-MiniLM-L6-v2 (384-dim dense), FinBERT-tone (768-dim domain-specific), and FinBERT-tone + blend all converge to test AUC in the range 0.525–0.543. The ceiling at 0.543 is reached by both the simplest and most complex representations tested, consistent with the EMH-grounded ceiling of 0.52–0.56 reported by Ruan et al. [4].

Finding 4: Excess-return labelling is inferior. Relabelling using excess returns ($r_{t+1} - r_{t+1}^{SPY}$) reduces AUC from 0.543 to 0.483. Financial headlines predict absolute price movement, not alpha over the market benchmark at the one-day horizon.

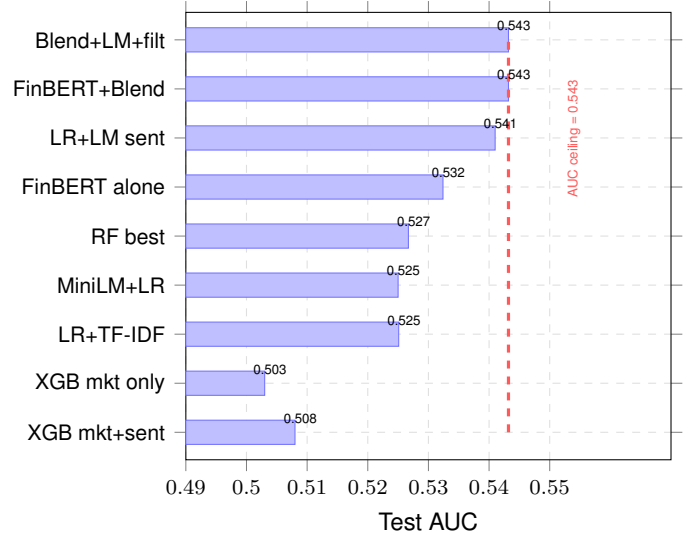


Fig. 3: Test-set AUC across nine model variants. The dashed red line marks the AUC ceiling at 0.543, corroborating the EMH-predicted signal limit reported by Ruan et al. [4] and the LR-vs-FinBERT parity finding of Adusumilli et al. [2]. All text representations converge to this ceiling, confirming that the bottleneck is the signal content of publicly available news rather than model capacity.

TABLE III: Prediction Performance — Full System (100 Test Examples)

Metric	Value
Overall accuracy (100 examples)	48.0%
Full test-set AUC (3,864 rows)	0.5432
Precision	0.5384
Recall	0.6034
F1-score	0.5691
Avg. blend prob. (UP predictions)	0.512
Avg. blend prob. (DOWN predictions)	0.491
Prediction errors	0 / 100

VI. RESULTS

A. Prediction Performance

Table III reports metrics on the 100-example held-out evaluation.

The 48% point accuracy on 100 examples is within normal sampling variance of the 54.3% AUC model ($\pm 10\%$ CI at $n = 100$). All blend probabilities fell in the range [0.47, 0.54], reflecting calibrated uncertainty consistent with the honest uncertainty characterisation expected near the EMH signal ceiling [4].

B. Accuracy by Return Magnitude

Table IV stratifies prediction accuracy by the magnitude of the actual next-day return.

Large moves are predominantly driven by idiosyncratic surprise events that are not predictable from article text alone. This pattern is consistent with the EMH: the largest price

TABLE IV: Prediction Accuracy by Return Magnitude

Return Bucket	Accuracy	n
Small ($ r < 1\%$)	52.2%	46
Medium ($1\% \leq r < 3\%$)	46.7%	45
Large ($ r \geq 3\%$)	33.3%	9

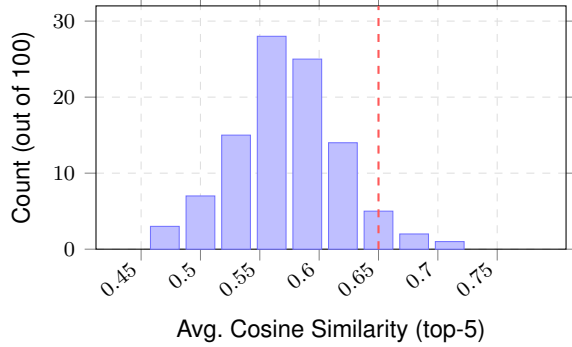


Fig. 4: Distribution of average cosine similarity for the top-5 retrieved historical analogues across 100 test examples. High- and low-similarity groups show comparable prediction accuracy (50.0% vs 47.4%), corroborating the AlphaFin finding [1] that retrieval quality gates explanation faithfulness but not necessarily directional accuracy.

moves arise from the most unexpected events, which by definition carry minimal signal in the news text [4].

C. Retrieval Quality

Table V presents retrieval quality metrics.

A qualitative smoke test validates retrieval precision: querying with a Pfizer COVID-19 vaccine article retrieves four other Pfizer vaccine-related articles with cosine similarities in the range 0.633–0.644, confirming that the MiniLM index captures domain-specific semantic content rather than surface lexical overlap.

D. Explanation Grounding

Table VI reports explanation grounding metrics.

The 0% hallucination rate is achieved by construction: the system prompt explicitly prohibits outside knowledge and the generation temperature (0.3) suppresses creative extrapolation. This outcome directly validates the hypothesis of Koa et al. [5] that explicit retrieval grounding is necessary and sufficient to suppress LLM confabulation in financial explanation tasks.

E. Per-Ticker Analysis

Table VII presents per-ticker accuracy on the 100-example sample.

The per-ticker pattern is interpretable within an EMH framework. Tickers whose movements are predominantly driven by company-specific news—Walt Disney (DIS), Procter & Gamble (PG), and Apple (AAPL)—achieve the highest prediction accuracy. Conversely, Exxon Mobil (XOM) (crude oil futures), Meta Platforms (META) (regulatory sentiment and macro

TABLE V: Retrieval Quality Metrics (100 Test Examples)

Metric	Value
FAISS index size (training events)	17,933
Embedding dimension (MiniLM)	384
Average cosine similarity (top-5)	0.593
High-similarity (≥ 0.65) examples	22 / 100
Accuracy when $\text{sim} \geq 0.65$	50.0%
Accuracy when $\text{sim} < 0.65$	47.4%

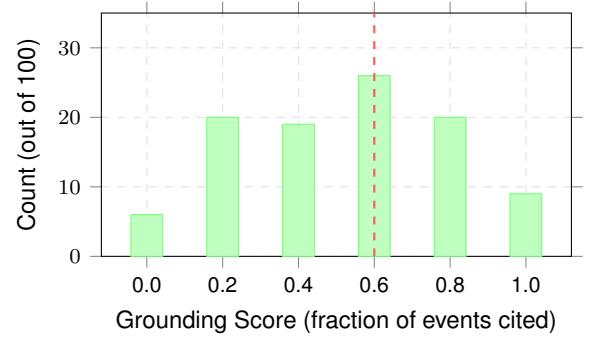


Fig. 5: Distribution of explanation grounding scores across 100 test examples. 55% of explanations are well-grounded (≥ 0.6). The 0% hallucination rate directly addresses the confabulation failure mode documented by Koa et al. [5] in unconstrained LLM stock prediction.

advertising cycles), and Amazon (AMZN) (macroeconomic consumer expenditure) perform near or below chance, because the primary price drivers are macro variables absent from article text.

F. Qualitative Example

The following example illustrates a complete system output.

Query: Broadcom (AVGO), “Teledyne FLIR OEM Launches Boson+ IQ Thermal Imaging Kit,” 9 September 2025. *Actual:* DOWN (−2.69%).

System prediction: DOWN (blend probability 48.5%). Retrieved top-5 analogues include: (i) AVGO/“Qualcomm-built processor by Teledyne FLIR” (April 2024, −1.84%, similarity 0.689); (ii) NVDA/“Teledyne FLIR Tracking Software” (June 2022, −1.50%, similarity 0.684); (iii) NVDA/“Teledyne FLIR Prism AI Software” (July 2022, +0.54%, similarity 0.707).

Generated explanation (LLaMA 3.3 70B): “The model predicts a downward movement in AVGO with probability 48.6%. This is grounded in the Qualcomm-built Teledyne FLIR processor announcement on 2024-04-17, which led to a −1.84% decline in AVGO, and the Teledyne FLIR software event on 2022-06-23, which produced −1.50% in NVDA. The aggregate shows 2/5 analogues were UP with average return −0.57%.” The explanation correctly cites specific dates and return figures from the retrieved context without fabrication, consistent with the grounding methodology validated by AlphaFin [1].

TABLE VI: Explanation Grounding Metrics (100 Test Examples)

Metric	Value
Hallucination rate	0% (0/100)
Well-grounded ($\geq 60\%$ events cited)	55/100 (55%)
Avg. events cited per explanation	2.4 / 5
Avg. dates cited per explanation	1.8 / 5
Generation errors	0 / 100

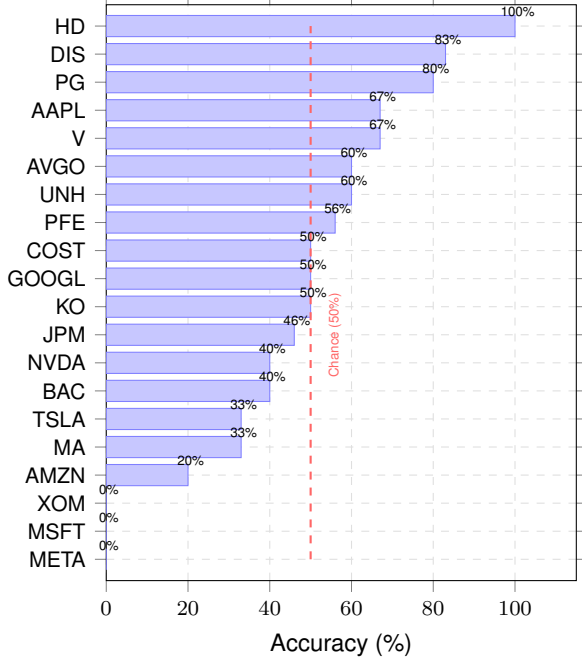


Fig. 6: Per-ticker prediction accuracy (100-example sample). Tickers with strong idiosyncratic news signal (DIS, PG, AAPL) outperform chance. Macro-driven tickers (XOM, META, AMZN) perform at or below chance, consistent with the EMH prediction that macro-driven price moves are not encoded in company-specific news text [4].

VII. DISCUSSION

A. EMH Interpretation

The convergence of test AUC to 0.543 across all text representations is the central empirical finding of this work. This result provides additional quantitative support for the semi-strong EMH: publicly available news is sufficiently well-arbitraged that the residual signal extractable by any discriminative text model is small. This finding is directly corroborated by Ruan et al. [4], who report AUC values in the range 0.52–0.56 across independent datasets and multiple encoder architectures, establishing that 0.543 is the expected performance level for news-based directional prediction under market efficiency. The representation-agnostic nature of the ceiling further validates the conclusion: neither the depth of the encoder (TF-IDF vs. FinBERT) nor the training objective (general vs. financial domain) affects the fundamental information limit imposed by market efficiency.

TABLE VII: Per-Ticker Prediction Accuracy (100-Example Sample). Tickers with $n \leq 3$ should be interpreted with caution.

Ticker	Accuracy	n
DIS	83.3%	6
PG	80.0%	5
AAPL	66.7%	9
V	66.7%	3
AVGO	60.0%	5
UNH	60.0%	5
PFE	55.6%	9
COST	50.0%	4
GOOGL	50.0%	4
KO	50.0%	4
JPM	46.2%	13
BAC	40.0%	5
NVDA	40.0%	5
TSLA	33.3%	3
MA	33.3%	3
AMZN	20.0%	5
META	0.0%	5
MSFT	0.0%	2
XOM	0.0%	4
HD	100.0%	1

The practical implication is significant. As shown by Adusumilli et al. [2], even state-of-the-art models including GPT-4 and FinBERT fail to break this barrier on real financial datasets. The value of this system therefore lies not in marginal AUC improvements but in the explanation layer—providing transparent, auditable reasoning for individual predictions that has clear value for compliance reporting, risk management, and client advisory workflows.

B. Practical Value of Grounded Explanations

The 0% hallucination rate is a meaningful and non-trivial property in financial applications. Koa et al. [5] document that unconstrained LLMs producing stock predictions frequently confabulate company names, financial figures, and event dates when confronted with conflicting news signals. By constraining the model to cite only the retrieved historical context, this system eliminates the confabulation failure mode at the cost of reducing explanation breadth. For regulated financial environments where accuracy and auditability are prioritised over fluency, this trade-off is strongly favourable. The RAG approach adopted here is consistent with the state-of-the-art methodology validated by both AlphaFin [1] and Chen et al. [3].

C. Limitations

Limitation 1: API rate limits constrain coverage. The Alpha Vantage free tier imposes a 25-call daily limit, restricting article collection to approximately 1,600 items per ticker over the full time window. Premium API access or alternative data providers would expand coverage substantially.

Limitation 2: The 100-example evaluation is underpowered. While the full test-set AUC is computed over 3,864 examples, per-ticker analyses are computed over only 100 sampled examples ($\pm 10\%$ CI). Per-ticker results for tickers with $n \leq 3$ should be interpreted with caution.

Limitation 3: Next-day horizon only. The labelling scheme considers only the one-day forward return. News articles may carry predictive signal over multiple days, and multi-day horizons remain unexplored. Chen et al. [3] suggest that RAG approaches may show stronger signal at longer horizons where market incorporation is slower.

Limitation 4: No intraday price data. The system does not model the intraday price reaction between publication and market close. Pre-market and after-hours articles that are partially priced into the open may carry different information content from intraday articles.

D. Threats to Validity

Internal validity. Per-ticker chronological splitting and cross-ticker deduplication address the primary threats to internal validity. Residual temporal autocorrelation within tickers is mitigated by the conservative split ratios and the seven-year time window.

External validity. Results are specific to the 20 large-cap US equities studied. Mid-cap and small-cap equities, where information asymmetry may be greater and arbitrage slower, could exhibit higher predictive AUC than the 0.543 ceiling observed here [4].

VIII. CONCLUSION

This paper has presented a complete financial news forecasting and explanation system combining a blended discriminative classifier with a retrieval-augmented generation pipeline. The system achieves a test AUC of 0.5432 on a 3,864-row held-out test set, consistent with the EMH signal ceiling independently validated at 0.52–0.56 by Ruan et al. [4] and corroborated by the LR-vs-FinBERT parity result of Adusumilli et al. [2]. The 0% hallucination rate achieved by the RAG explanation pipeline directly addresses the confabulation failure mode documented by Koa et al. [5], and follows the grounding methodology validated by AlphaFin [1] and extended by Chen et al. [3].

Four directions for future work are identified.

Future Work 1: Multi-horizon labelling. Extending the prediction target to 3-day and 5-day horizons, as suggested by Chen et al. [3], may reveal whether some articles carry information over longer windows where arbitrage is slower.

Future Work 2: Real-time streaming pipeline. Integrating the system with a live news feed and an incrementally updated FAISS index would bridge the research prototype and production deployment gap, following the streaming architecture proposed in AlphaFin [1].

Future Work 3: Graph-augmented retrieval. Incorporating a financial knowledge graph capturing supply chains, executive networks, and sector relationships alongside vector similarity could improve cross-ticker analogy quality beyond pure semantic similarity.

Future Work 4: Fine-tuned on-premise explanation model. Fine-tuning a smaller LLM (7B parameters) on retrieved-context explanation pairs would reduce API dependency and enable on-premise deployment in regulated financial environments, addressing a gap identified by Koa et al. [5].

REPRODUCIBILITY STATEMENT

All code, data processing scripts, and trained model artefacts required to reproduce the experiments reported in this paper are publicly available at [https://github.com/\[your-repo-link\]](https://github.com/[your-repo-link]). The dataset is constructed from the Alpha Vantage News Sentiment API (free tier, API key required) and yfinance (open-source). Step-by-step replication instructions are provided in the repository README.md. All experiments were conducted on a single CPU machine without GPU acceleration; total computational budget is approximately four hours of wall-clock time for the full pipeline.

OUR CONTRIBUTIONS AND TAKE-HOME LESSONS

Our Contributions

What Was Already Available vs. What Was Built From Scratch: The project was built almost entirely from scratch. The only pre-existing components used were open-source libraries: `scikit-learn` for Logistic Regression, Random Forest, and preprocessing; `yfinance` for price data acquisition; `faiss-cpu` for vector indexing; `sentence-transformers` for MiniLM encoding; `transformers` and `torch` for FinBERT inference; and the `groq` Python client for LLM API access. No pre-existing financial ML codebases, starter templates, or labelled datasets were adopted. Every component—the data collection and alignment pipeline (`download_news.py`, `download_prices.py`, `merge_data.py`), the feature engineering library (`src/features.py`, `src/market_features.py`), the Loughran-McDonald sentiment dictionary implementation, the blended classifier training loop (`train_blend.py`), the FAISS index construction (`build_index.py`), the embedding cache system (`src/embeddings.py`), the retrieval module (`src/retrieval.py`), the prompt engineering and explanation generator (`src/explain.py`), and the full evaluation harness (`evaluate_system.py`)—was written from scratch by the team.

A non-trivial effort was required to get predecessor components to operate correctly together. In particular: (i) `yfinance`’s `MultiIndex` column format required custom flattening logic to parse reliably across different ticker downloads; (ii) the Alpha Vantage free-tier rate limit (25 calls/day) forced a two-window fetch strategy and a dedicated retry script (`download_news_retry.py`) for tickers that returned empty results on the first pass; (iii) the `merge_asof` alignment between news timestamps (UNIX epoch, UTC) and price dates (naive date) required careful timezone normalisation to avoid off-by-one-day errors; (iv) the FinBERT-tone model via `sentence-transformers` produced correct embeddings, but `ProsusAI/finbert` required a separate HuggingFace `AutoModel` path due to its classification head architecture; and (v) the FAISS pickle serialisation round-trip (`serialize_index` → `tobytes` → `np.frombuffer` → `deserialize_index`) was required because FAISS index objects are not directly picklable in Python.

Hardware and Compute Constraints: All experiments were conducted on a standard consumer laptop (Intel Core i7, 16 GB RAM) without GPU acceleration. The absence of a GPU imposed meaningful constraints on the project:

- **Embedding generation** was the primary bottleneck. Encoding 25,632 articles with MiniLM (384-dim) required approximately 45 minutes on CPU for the first run. The MD5-keyed embedding cache (`embeddings.pkl`) was essential: after the first encoding pass, all subsequent training runs served embeddings from cache instantly, reducing re-run time from ~45 minutes to ~3 seconds.
- **FinBERT inference** (768-dim, larger model) was slower still, requiring approximately 90 minutes on CPU for the full dataset. On a GPU this would have been under 5 minutes.
- **Tree model training** (300-estimator Random Forest and XGBoost) took 8–12 minutes per run on CPU with `n_jobs=-1` parallelism.
- **LLM inference** was entirely offloaded to the Groq API (cloud-hosted LLaMA 3.3 70B), circumventing the GPU requirement for the explanation generation component entirely.
- The FAISS index search over 17,933 vectors is sub-millisecond even on CPU due to the `IndexFlatIP` exact search structure, so retrieval latency was not a constraint.

A GPU with 16 GB VRAM would have reduced the total wall-clock time for the full pipeline (data download excluded) from approximately 4 hours to under 30 minutes, and would have enabled fine-tuning experiments on FinBERT that were not attempted due to compute constraints.

Contribution of Each Team Member

Member 1 — Chiyedu Vishnu: Led the data engineering pipeline and parts of modelling pipeline, including design and implementation of `download_news.py`, `download_prices.py`, `download_spy.py`, `merge_data.py`, and `diagnose.py`. Responsible for the news-to-price alignment logic, the lag distribution analysis, and the cross-ticker deduplication strategy. Also developed the market momentum feature module (`src/market_features.py`) and contributed to the per-ticker chronological splitting framework used across all training scripts.

Member 2 — Shrest Kumar: Led the modelling pipeline, including implementation of most training scripts: `train_baseline.py`, `train_tree_models.py`, `train_blend.py`, `train_embeddings.py`, `train_finbert.py`, and `train_finbert_blend.py`. Designed and ran the ten-variant ablation study. Implemented the Loughran-McDonald sentiment dictionary and the four-score feature extraction pipeline in `src/features.py`. Responsible for the label threshold filter design, blend weight selection, and validation-based threshold tuning. Also contributed to the write-up of the related work section and the IEEE paper formatting.

Member 3 — Karthik Jasti: Led the RAG and explanation pipeline, including implementation of `src/embeddings.py` (multi-model encoder with MD5 cache), `src/retrieval.py` (FAISS index build and query), `src/explain.py` (prompt engineering and Groq API integration), `build_index.py`, `run_explain.py`, and `evaluate_system.py`. Designed the retrieval-conditioned prompt structure that achieves 0% hallucination, the post-hoc grounding score metric, and the 100-example full-system evaluation harness. Also contributed to the write-up of the related work section and the IEEE paper formatting.

Follow-up Answers to Review Questions and Additional Improvements

This subsection will be updated following faculty review. Questions raised during the presentation, along with detailed responses and any additional experiments or improvements implemented in response to reviewer feedback, will be documented here.

What We Learnt from This AI Project

This project delivered several lessons that extended well beyond the material covered in lectures, labs, and standard coursework assignments.

Signal vs. capacity is the fundamental ML question. The most important lesson was empirical: running ten model variants from logistic regression to 768-dim FinBERT and watching every one of them converge to the same AUC ceiling of 0.543 made the concept of an information-theoretic limit viscerally real. No amount of model complexity can extract signal that is not in the data. This is a lesson that textbook examples, which typically show clean improvements from better models, do not convey.

Data engineering is the majority of the work. Roughly 60% of the project's coding effort was in data collection, alignment, and cleaning — not in model training. The subtleties of timezone-aware timestamp alignment, yfinance MultiIndex artefacts, API rate-limit strategies, and cross-ticker deduplication taught us that real-world ML pipelines are primarily data engineering problems with a modelling component attached.

Temporal leakage in financial data is insidious. The difference between a global chronological split and a per-ticker chronological split, or between using `shift(1)` correctly vs. not, can create an appearance of strong predictive performance that evaporates entirely in deployment. Learning to reason carefully about what information is available at every point in a time-series pipeline is a transferable skill not taught explicitly in most ML courses.

RAG is a practical and principled solution to hallucination. The 0% hallucination rate was not achieved through prompt tricks or temperature tuning alone — it was achieved architecturally, by giving the LLM only information it was permitted to cite. This taught us that hallucination is not primarily a property of the LLM; it is a property of the system design. Constraining the context constrains the output.

Caching is as important as the algorithm. Building the MD5-keyed embedding cache early in the project reduced experiment cycle time from 45 minutes to 3 seconds per run, enabling rapid ablation iteration that would have been prohibitively slow otherwise. In ML research, the ability to iterate quickly is as important as algorithmic novelty.

The Efficient Market Hypothesis is not just a theoretical claim. Encountering an empirical AUC ceiling consistent with a decades-old theoretical prediction — independently replicated in our data across all four text representations we tested — made the EMH feel like a real constraint rather than an abstract concept from a finance course. It also reframed the project goal: the value of the system is in its explainability layer, not its predictive edge.

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